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Disciplinary and Interdisciplinary Science Education Research (2026) 8:6
https://doi.org/10.1186/s43031-026-00154-2

Disciplinary and Interdisciplinary Science Education Research

EDITORIAL **Open Access**

Artificial intelligence in science education: global insights and future directions

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Introduction

Artificial Intelligence (AI) now stands at the forefront of transformative change across global society, reshaping how we approach science education at every level. The proliferation of AI applications, from machine learning algorithms that analyze student responses to large language models (LLMs) that generate instructional materials, has created unprecedented opportunities to enhance how students learn and how teachers teach. These innovations have transformed the landscape of science education by enabling new forms of assessment, personalized feedback, and adaptive learning environments. Yet, these same developments also raise pressing questions about ethics, equity, transparency, and cultural responsiveness, questions that go to the heart of how learners produce, share, and use knowledge in classrooms around the world.

In this evolving landscape, AI is increasingly conceptualized not merely as a technological tool but as a cognitive partner (Shin et al., 2025), a system that can extend human reasoning, interpretation, and creativity in educational contexts. Rather than replacing teachers or learners, AI can augment their capacity to analyze data, explore scientific phenomena, and engage in reflective inquiry. This framing shifts the conversation from *automation* to *augmentation*, highlighting the collaborative potential between human and machine cognition. Meaningful integration of AI requires rethinking epistemic relationships in learning, seeing AI as a co-participant that co-designs assessments for measuring students' usable knowledge in science education (Li et al., 2025). The global shift in science education toward helping learners *use* knowledge rather than merely recall it (Pellegrino & Hilton, 2012) makes the exploration of AI's role particularly timely. In the past decade, scholars have leveraged machine learning to automatically score student artifacts (e.g., Liu et al., 2016; Zhai et al., 2022), generate formative feedback for teachers (e.g., He et al., 2024b), and support students' model-based reasoning and explanation construction (e.g., He et al., 2024a; Zhu et al., 2020). These developments align closely with the core vision of the *Next Generation Science Standards* (NGSS, NGSS Lead States, 2013) and similar frameworks worldwide (e.g., Chinese Ministry of Education, 2017; Organisation for Economic Co-operation and Development, 2023), which emphasize the integration of disciplinary and interdisciplinary core ideas and scientific and engineering practices. However, the rapid acceleration of generative AI, especially systems such as ChatGPT, Gemini, and Claude, has dramatically expanded both the possibilities and uncertainties surrounding AI's use in STEM Education.

Across contexts, teachers and researchers are grappling with several fundamental questions: How can AI be used to promote usable knowledge? In what ways can teachers leverage AI to interpret student thinking and provide timely, actionable, and constructive feedback? How do we ensure that AI-enhanced learning environments remain equitable, transparent, and aligned with human values? And perhaps most importantly, how do we prepare the next generation of teachers and students to engage with AI critically and responsibly? These questions demand not only technical innovation but also

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pedagogical, ethical, and cultural reflection. Science education provides a vibrant and challenging context for examining these issues. Learning science involves grappling with uncertainty, constructing and revising models, and using evidence to explain natural phenomena (NRC, 2012). All processes that align with the reasoning capabilities AI technologies now emulate. Yet, effective science teaching also depends on empathy, cultural attunement, and professional judgment, dimensions of practice not easily automated. (e.g., Avraamidou, 2024; Li et al., 2023).

The growing body of research represented in this special issue reveals how AI's affordances and constraints intersect with these human dimensions of teaching and learning. This special issue brings together nine studies from five countries across three continents: Asia, Europe, and North America. These papers collectively examine how researchers designed, implemented, and interpreted AI to support science teaching and learning in formal and informal settings. The contributions span a wide range of topics, from preparing teachers to navigate AI-rich classrooms and using LLMs for reflective practice, to advancing AI-based learning analytics and cultivating critical AI literacies within communities. Together, they reflect a rapidly evolving research frontier that seeks to balance technological innovation with educational purpose and ethical responsibility. Across these contributions, four interconnected themes capture the field's current progress and future directions. The first, *Reimagining Teacher Preparation for an AI-Infused Future*, highlights how teacher education programs are addressing new competencies, ethical considerations, and pedagogical adaptation in AI-mediated contexts. The second, *Using AI to Amplify Teacher Knowledge, Reflection, and Practice*, illustrates how AI can support teacher learning, reflection, and analysis of student thinking. The third, *Advancing Student Learning Through AI-Driven Assessment and Learning Analytics*, focuses on using AI to reveal learning trajectories and conceptual growth. Finally, *Fostering Critical and Community-Centered AI Literacies* expands the conversation beyond classrooms, emphasizing social, cultural, and ethical engagement with AI in science education. Together, these studies reveal how AI integration invites educators to reconsider the evolving relationships among pedagogy, technology, and human learning. The contributions in this special issue illuminate how researchers and educators around the world are beginning to navigate this complex relationship, offering visions of AI-enabled science education that are more equitable, reflective, and future-oriented.

Overview of the four themes

The studies in this special issue form a varied yet interconnected body of work that reflects global developments in AI for science education. Together, they reveal how researchers are tackling shared obstacles while adapting to specific localities. Four key themes emerge, each demonstrating both opportunities and tensions in the field's growth. These themes show progress in embedding AI as an integral part of science teaching, learning, and professional development (NRC, 2006). The included research covers teacher preparation, professional reflection, learning analytics, and community-based AI literacy. While each study addresses a unique context, the overarching themes present both advances and outstanding questions that define human–AI collaboration in contemporary science education.

Theme 1. Reimagining teacher preparation for an AI-infused future

This theme emphasizes the foundational role of teacher education in shaping equitable and effective AI integration in science classrooms. Across two studies from the Philippines and Indonesia, the authors explore how pre-service teachers perceive, experience, and learn to work with AI. Vera et al. examine pre-service science teachers' competencies for an AI-enhanced future, revealing an urgent need to connect aspirational skills, such as creativity, ethical reasoning, and adaptability, with concrete pedagogical strategies. Arini et al. focus on AI integration to support multilingual learners, demonstrating teachers' recognition of AI's equity potential alongside their lack of structured AI training. These studies highlight that preparing teachers for AI-mediated education is not solely about building technical skills. It is about cultivating ethical awareness, pedagogical adaptability, and critical reflection. They underscore the importance of embedding AI literacy and data ethics into teacher education programs, ensuring that teachers emerge as thoughtful designers and evaluators of AI tools rather than passive adopters. We recommend that future work should explore longitudinal and cross-cultural models of AI-integrated teacher preparation, examining how teachers' beliefs and practices evolve as AI becomes part of their daily instructional routines. There is also a pressing need to investigate how AI can support equitable participation among teachers and students in linguistically and culturally diverse contexts.

Theme 2. Using AI to amplify teacher knowledge, reflection, and practice

This theme focuses on how AI can expand teachers' professional learning and interpretive capacity. Rachmatullah et al. investigate the use of large language models (LLMs) to assess science teachers' pedagogical content knowledge (PCK), finding that AI systems can achieve reliability comparable to that of human raters and open scalable possibilities for teacher knowledge research. Sorge et al. demonstrate how fine-tuned LLMs can provide individualized feedback on pre-service physics teachers' written reflections, emphasizing the importance of contextual specificity to improve model accuracy. Jin et al. explore how pre-service elementary teachers collaborate with ChatGPT to analyze student responses, revealing how AI can prompt more sophisticated reasoning but also challenge teachers' ability to interpret lengthy, detail-heavy outputs. Collectively, these studies demonstrate that AI can serve as a mirror and mentor for teachers, helping them analyze student thinking, articulate pedagogical reasoning, and develop reflective habits. AI-enabled feedback systems and analytic tools have the potential to democratize access to mentoring and support in teacher education programs, particularly where expert feedback is scarce. Moving forward, researchers should examine how to embed AI-supported reflection tools into teacher professional development ecosystems and how these systems can adapt to teachers' evolving goals and cultural contexts. There is also a need to interrogate the epistemic alignment between AI-generated feedback and teachers' pedagogical values, ensuring that AI amplifies rather than constrains professional judgment.

Theme 3. Advancing student learning through AI-driven assessment and learning analytics

The third theme highlights how AI can help uncover students' conceptual development and learning trajectories in new and powerful ways. Wang et al. integrate ontological frameworks with pretrained BERT models to classify students' scientific explanations, demonstrating how domain-specific ontologies can enhance model interpretability and accuracy. Domenichini et al. apply AI-driven network analysis to map students' evolving understanding of energy concepts, capturing non-linear learning pathways that traditional assessments often overlook. Wyrwich et al. employ unsupervised machine learning (i.e., k-means longitudinal analysis) to characterize students' energy learning trajectories, represented by their energy knowledge network coherence. Together, these studies show that AI-based analytics can transform assessment from a static measure of knowledge to a dynamic representation of learning in progress. These contributions highlight AI's potential to advance formative assessment and adaptive instruction, offering educators granular insights into how students connect, reorganize, and extend their ideas. By visualizing individual and group learning trajectories, AI-based systems can make complex patterns of reasoning more visible and actionable for teachers and researchers. Future research should investigate how teachers interpret and utilize AI-generated analytics to inform classroom instruction. Moreover, the field must continue to address the ethical, interpretive, and validity challenges associated with AI in assessment, ensuring that these powerful tools remain transparent, trustworthy, and aligned with educational theory rather than relying solely on data-driven logics.

Theme 4. Fostering critical and community-centered AI literacies

The final theme extends beyond classroom-based teaching to the broader civic and ethical dimensions of AI in STEM education. Akgün et al. propose a community-centered, participatory approach to developing critical AI literacy programs for K–8 learners, co-designing curricular modules with students and public librarians. Their design process exemplifies how local communities can play a vital role in shaping young learners' understanding and engagement with AI, emphasizing social responsibility, ethical reflection, and cultural relevance. This work reframes AI literacy as a collective and ethical practice rather than a technical competency. By engaging communities as co-designers, this approach broadens the purpose of science education from empowering learners with technical skills to critically examining how AI systems shape society, knowledge, and equity. Future research can build on this work by developing scalable models of community participation in AI education that cross formal and informal learning contexts. There is also a growing need to examine how critical AI literacy can be integrated into national curricula and teacher preparation programs, helping all learners, not only future data scientists, become informed participants in an AI-driven world.

Conclusions and future directions

Across these four themes, the studies collectively reveal a field in transformation. Science education is moving toward an era in which AI is not a supplement but a structural component of learning ecosystems, dynamically interacting with teachers' expertise, students' agency, and societal values (Avraamidou, 2024; Cooper & Klymkowsky, 2024; He et al., 2024b). These contributions mark an essential step in building a more human-centered, globally informed vision of AI in science education, one that balances innovation with responsibility and curiosity with care. Artificial intelligence is not merely a technological advance; it is a catalyst for reimagining what it means to learn, teach, and conduct science education research in the 21st century. The studies in this special issue collectively show that AI's potential lies not in automating human work (e.g., Liu et al., 2016; Zhai et al., 2022) but in deepening the relational, reflective, and intellectual dimensions of science education. Together, they illustrate a transition from *AI for efficiency* toward *AI for equity, inquiry, and cognition*, a shift that foregrounds collaboration between human and artificial intelligence to advance the goals of usable knowledge, epistemic agency, and social responsibility.

A *central insight* cutting across the contributions is **the importance of teacher agency**, **reflection**, **and preparedness**. Teachers occupy a critical position at the interface between AI innovation and classroom practice. Across contexts, from pre-service programs in the Global South to professional learning environments in Europe and the United States, teachers are eager yet uncertain about how to integrate AI meaningfully and ethically. The studies emphasize that technical competence alone is insufficient. Teachers must develop a reflective understanding of AI's possibilities and limitations, along with the ethical and pedagogical frameworks needed to make informed decisions. Building such capacity requires embedded, practice-based professional learning in which teachers co-design, critique, reflect upon, and adapt AI tools within authentic instructional contexts. The *second insight* focuses on **AI's epistemic and methodological contributions.** Several papers demonstrate how AI can extend, not replace, human interpretation in research and practice. Through large language models, network analyses, and ontology-based classifiers, AI enables researchers and educators to see learning in new ways: as dynamic, multi-layered, and evolving. These systems offer a lens into students' conceptual trajectories and teachers' professional reasoning, revealing patterns of growth and struggle previously invisible in traditional assessment frameworks. Yet, the interpretive power of AI also demands caution. The field must continue to interrogate the epistemological assumptions underlying AI models. How to represent data, what counts as evidence, and whose ways of knowing are privileged. This reflection is essential if AI is to enhance, rather than distort, educational meaning-making. The *third insight* that unites these studies is **the tension between innovation and equity.** AI has the potential to personalize learning, offer multilingual support, and reduce barriers to participation. However, as the authors in this special issue caution, and we agree with caution, it can also reproduce existing inequities through biased datasets, opaque algorithms, and cultural misalignment. Equitable AI integration must therefore extend beyond access and functionality to include questions of representation, fairness, and agency. This requires collaboration between educators, computer scientists, policymakers, and communities to design AI systems that are inclusive, transparent, and responsive to diverse learners and local educational needs. Finally, the studies collectively call for a humanistic and participatory vision of AI in science education. The work of Akgün and colleagues exemplifies how critical and community-centered AI literacies can empower learners to engage not only with AI technologies but also with the ethical and societal questions they raise. Across contexts, the future of AI in science education depends on positioning learners, teachers, and communities as co-creators of intelligent systems, as participants who question, design, and guide AI to serve educational and civic purposes. This participatory turn moves the field toward a vision of AI as part of a broader socio-technical ecosystem that reflects shared human values and educational aspirations.

Taken together, the nine articles from five countries included in this special issue offer a globally situated yet conceptually coherent perspective on the opportunities and challenges of using AI in science education. While each study is grounded in a specific educational context, collectively they highlight shared concerns around teacher agency, student learning, equity, and the ethical integration of AI into science teaching and learning. These crosscutting insights provide a foundation for identifying broader questions that extend beyond individual studies and point toward future directions for the field. Looking ahead, several crosscutting questions warrant continued inquiry: (1) *How can we ensure that AI augments, rather than replaces, human judgment and relational teaching? (2) In what ways can AI-generated analytics and feedback be made interpretable and pedagogically meaningful to teachers and learners? (3) How can teacher education systems worldwide integrate AI literacy and ethics while respecting local cultures and resources? (4) What frameworks are needed to evaluate AI's long-term impact on student learning, identity, and equity?* Responding to these questions requires sustained collaboration across disciplines, nations, and professional communities. As the contributions in this special issue demonstrate, the integration of AI in science education is not simply an innovation challenge. It is an epistemic, ethical, and cultural project that spans across cultures, continents, and people. The next generation of researchers must therefore aim to design AI systems that are transparent, trustworthy, and transformative, and that not only analyze learning but also honor the human curiosity and creativity at the heart of science education.

Author Contribution

PH wrote the draft, JK reviewed the draft; all authors agreed on the content of this editorial.

Declarations

Competing interests

None.

Published online: 02 February 2026

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